

GHz. Simple math would tell you that amounts to 5000+ GHz or more than 5 trillion instructions per second! That's a staggering number. Considering that a lot of tasks can be done by other hardware chips, the processing power of a graphic cards is too high. However, they work on a SIMD pattern and have to work in parallel. Game design principals and standards (like DirectX and Metal) ensure that the processing patterns utilize this architecture to its max.

3. **Types of memory:** You might have heard some people saying something like "XYZ graphics card has 6 GB of RAM". Well, such a statement would be wrong and if you're guilty of it yourself, please correct yourself. Graphics cards do not have normal RAM. They instead have GDDR RAM. The extra 'G' stands for 'graphics' (quite obviously). Graphics RAM is divided into multiple chunks. There are just as many divisions as the number of processors available. When the graphics card is running, all its processors try to access the memory. Each processor can simultaneously access the memory from a division of memory. So if there are 2048 processors and each wants to read a 1 MB of data, the GDDR RAM will deliver 1MB to the 2048 processors in parallel (a total of 2 GB). This parallel access pattern is what sets GDDRRAM apart from normal RAM which works serially (processing one request after another). An upcoming type is HBM or High Bandwidth Memory.

There is another type of memory available to a graphics card which is called 'Texture memory'. This type of memory allows the processor to read data in "surrounding memory units" of an address much faster.

Which graphics card is good for me?

If gaming is fun, graphics cards are the humour. They are also ridiculous when it comes to sheer computational output with over 12 Billion (with a capital 'B') transistors in the latest models. While those might be very costly, there is one for everyone.

Basic Rig

If you are building a basic PC, we do not think you have many options (or need) here. It would be best to settle with the one built into your processor.

Entry Level Gaming

With 2 GB of GDDR5 memory, a 1345 MHz clock and 640 processing cores, the Zotac GTX 1050 is a good piece to make some noise in the box.